



Power BI & DAX Learning Journey



Introduction

This document represents my learning journey in Power BI and DAX (Data Analysis Expressions).

It includes all the topics, concepts, and hands-on practice I have completed during my coursework.

The goal of this learning phase was to build a strong foundation in data analytics before moving on to real-world projects.



Tools & Technologies Used

- Power BI
- Microsoft Excel / CSV datasets
- DAX (Data Analysis Expressions)



Topics Covered

♦ Module 2: Introduction to Data

- Understanding datasets
- Types of data (structured / unstructured)
- Data loading in Power BI
- Basic data exploration

♦ Module 3: Data Cleaning

- Handling missing values
- Removing duplicates
- Data formatting
- Data transformation using Power Query

♦ Module 4: Data Transformation

- Column operations
- Filtering and sorting data
- Splitting and merging columns
- Creating calculated columns

♦ **Module 5: Data Modeling Basics**

- Relationships between tables
- Primary and foreign keys
- Star schema basics
- Data model design

♦ **Module 6: Introduction to DAX**

- Understanding DAX syntax
- Calculated columns vs measures
- Basic DAX functions:
 - SUM
 - COUNT
 - AVERAGE

♦ **Module 7: Intermediate DAX**

- Logical functions (IF, SWITCH)
- Aggregation functions
- Filtering concepts
- Context (Row Context vs Filter Context)

♦ **Module 8: Advanced DAX Concepts**

- CALCULATE function
- ALL and FILTER functions
- Time-based calculations (basic level)
- Creating dynamic measures

♦ **Module 9: Practice & Assignments**

- Solved multiple datasets
- Applied DAX functions in real scenarios
- Built basic reports using Power BI
- Improved problem-solving skills



Practical Work Done

- Worked with multiple uncleaned datasets
- Performed data cleaning and transformation
- Created calculated columns and measures
- Practiced DAX functions on real datasets
- Built basic Power BI reports



Key Learning Outcomes

- Strong understanding of DAX fundamentals
- Ability to clean and transform data
- Knowledge of data modeling basics
- Hands-on experience with Power BI
- Improved analytical thinking



Conclusion

This learning journey helped me build a strong foundation in Power BI and DAX.

I am now prepared to work on real-world data analytics projects such as sales dashboards, business reports, and data-driven decision-making solutions.



Future Scope

- Build end-to-end Power BI projects
- Work on real-world datasets
- Create advanced dashboards
- Apply DAX in business use cases



Note

This document represents my initial learning phase and continuous improvement in the field of Data Analytics.